

# Aadhaar Card Layouts for OCR Analysis

## **1 Aadhaar Card Layout Variations**

Aadhaar is the national identity document issued by the Unique Identification Authority of India (UIDAI). During OCR benchmarking, multiple document layouts must be considered because they differ in structure, printing quality, background, acquisition conditions and privacy features. Although the semantic content remains largely identical, these variations influence text detection, layout analysis and information extraction.

The e-Aadhaar document is a printable PDF containing bilingual text, QR codes, identity information and detachable card sections. It represents the canonical layout used for OCR evaluation.

### Aadhaar Card Layout

[illegible]

## 1.2 PVC Aadhaar Card



Figure 2: Official PVC Aadhaar card (front and back).

The PVC Aadhaar card is the standard wallet-sized identity card issued by UIDAI. Its compact design requires accurate localization of small text fields and QR codes.

### 1.3 Printed/Laminated Aadhaar

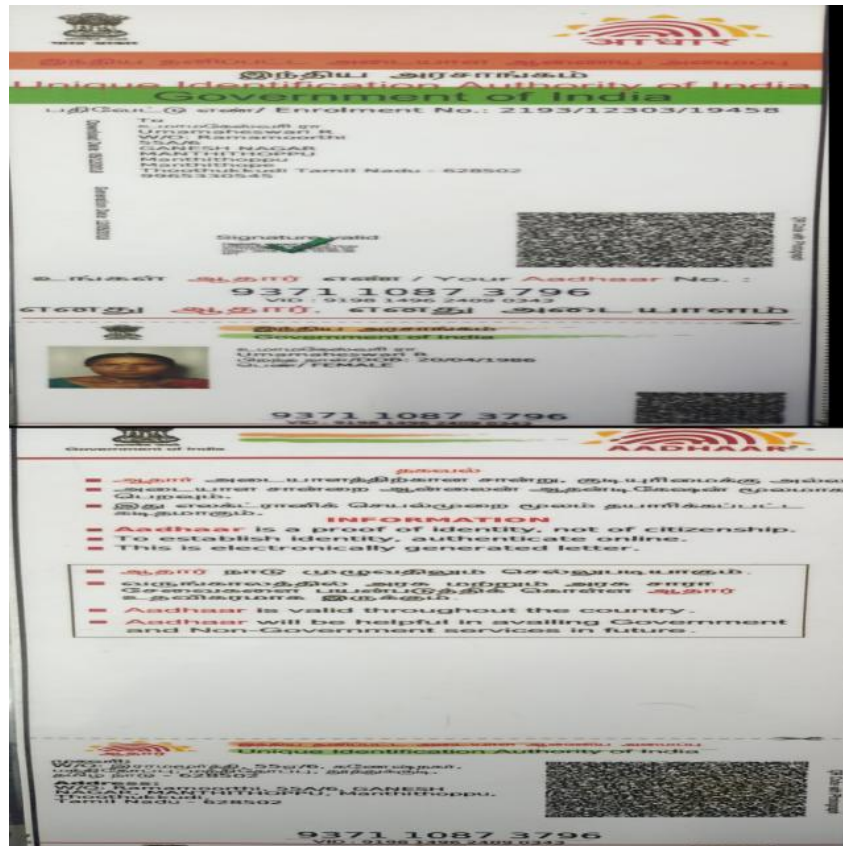


Figure 3: Printed or laminated Aadhaar document.

Printed copies introduce scaling differences, folds and lamination glare, making OCR more challenging.

## 1.4 Masked Aadhaar



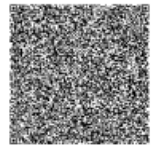

**भारत सरकार**  
Government of India

**भारतीय विशिष्ट पहचान प्राधिकरण**  
Unique Identification Authority of India

नामांकन क्रम/ Enrolment No.: 0013/09036/07772

To  
अदित्य कुमार  
Aditya Kumar  
C/O Mukesh Kumar,  
Alok Bhawan Gali,  
Near Yogipur Shivmandir,  
Yogipur,  
VTC: Kankarbagh,  
PO: Lohia Nagar,  
District: Patna,  
State: Bihar,  
PIN Code: 800020  
Mobile: 7294069111

Signature Verified  
Digitally signed by Aditya Kumar  
DN: cn=Aditya Kumar, o=Government of India  
Date: 2018.03.30 11:54:07  
+05'30'



आपका **आधार** क्रमांक / Your **Aadhaar** No. :  
**XXXX XXXX 3131**  
VID: 9124 0336 2934 4199

**मेरा आधार, मेरी पहचान**




**सूचना / INFORMATION**

- **आधार** पहचान का प्रमाण है, नागरिकता या जन्मतिथि का नहीं। जन्मतिथि आधार नंबर धारक द्वारा प्रस्तुत सचुना और विनियमों में निर्दिष्ट जन्मतिथि के प्रमाण के दस्तावेज पर आधारित है।
- इस **आधार** पर को बुकईडीआई द्वारा नियुक्त प्रमाणिकरण एजेंसी के जरिए ऑनलाइन प्रमाणिकरण के द्वारा सत्यापित किया जाता चाहिए या ऐप स्टोर में उपलब्ध एमआधार या आधार ब्यूझर कोड स्कैनर ऐप से ब्यूझर कोड को स्कैन करके या [www.uidai.gov.in](http://www.uidai.gov.in) पर उपलब्ध सुरक्षित ब्यूझर कोड रीडर का उपयोग करके सत्यापित किया जाना चाहिए।
- **आधार** विशिष्ट और सुरक्षित है।
- पहचान और पते के समर्थन में दस्तावेजों को **आधार** के लिए नामांकन की तारीख से प्रत्येक 10 वर्ष में कम से कम एक बार **आधार** में अपडेट करना चाहिए।
- **आधार** विभिन्न सरकारी और गैर-सरकारी पर्यटन/सेवाओं का जाल सेने में सहायता करता है।
- **आधार** में अपना मोबाइल नंबर और ईमेल आईडी अपडेट रखें।
- **आधार** सेवाओं का लाभ लेने के लिए एमआधार ऐप डाउनलोड करें।
- **आधार/बीयोमेट्रिक्स** का उपयोग न करने के समय सुरक्षा सुनिश्चित करने के लिए **आधार/बीयोमेट्रिक्स** लॉक/अनलॉक सुविधा का उपयोग करें।
- **आधार** की मांग करने वाले सहमति लेने के लिए बाध्य हैं।
- **Aadhaar** is proof of identity, not of citizenship or date of birth (DOB). DOB is based on information supported by proof of DOB document specified in Aadhaar number holder.
- **Aadhaar** can be verified through either online authentication or offline appointed authentication agency or QR code scanning using mAadhaar or Aadhaar QR Scanner app available in app stores or using secure QR code reader app available on [www.uidai.gov.in](http://www.uidai.gov.in).
- **Aadhaar** is unique and secure.
- Documents to support identity and address should be updated in **Aadhaar** after every 10 years from date of enrolment for **Aadhaar**.
- **Aadhaar** helps you avail of various Government and Non-Government benefits/services.
- Keep your mobile number and email id updated in **Aadhaar**.
- Download mAadhaar app to avail of **Aadhaar** services.
- Use the feature of Lock/Unlock **Aadhaar**/biometrics to ensure security when not using **Aadhaar**/biometrics.
- Entities seeking **Aadhaar** are obligated to seek consent.

**aadhaarinfo** can be verified through either online authentication or offline appointed authentication agency or QR code scanning using mAadhaar or Aadhaar QR Scanner app available in app stores or using secure QR code reader app available on [www.uidai.gov.in](http://www.uidai.gov.in).

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Unique Identification Authority of India

Aadhaar no. is masked: 19032017



अदित्य कुमार  
Aditya Kumar  
जन्म तिथि: 30/03/2005  
पुरुष/ MALE

आधार पहचान का प्रमाण है, नागरिकता या जन्मतिथि का नहीं। इसका उपयोग सचुना (ऑनलाइन प्रमाणिकरण, या ब्यूझर कोड/ऑनलाइन एकाएकरी कोड स्कैनर) के साथ किया जाना चाहिए।  
**Aadhaar is proof of identity, not of citizenship or date of birth. It should be used with verification (online authentication, or scanning of QR code / offline XMI).**

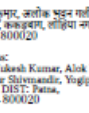
**XXXX XXXX 3131**

**मेरा आधार, मेरी पहचान**




**भारतीय विशिष्ट पहचान प्राधिकरण**  
Unique Identification Authority of India

Aadhaar no. is masked: 20092016



अदित्य कुमार, अलोक भवन गली, योगीपुर शिवमंदिर के पास,  
योगीपुर कोडबग, लोहिया नगर, पटना,  
पिन - 800020

Address:  
C/O Mukesh Kumar, Alok Bhawan Gali, Near  
Yogipur Shivmandir, Yogipur, Kankarbagh, PO: Lohia  
Nagar, DIST: Patna,  
Bihar - 800020

**XXXX XXXX 3131**  
VID: 9124 0336 2934 4199

**मेरा आधार, मेरी पहचान**

1547 | [help@uidai.gov.in](mailto:help@uidai.gov.in) | [www.uidai.gov.in](http://www.uidai.gov.in)

Figure 4: Masked Aadhaar layout.

Masked Aadhaar replaces the first eight digits of the Aadhaar number with “XXXX XXXX” while preserving the remaining document structure. OCR systems must correctly identify masked characters without confusing them with numeric digits.

## 1.5 Real Camera-Captured Aadhaar

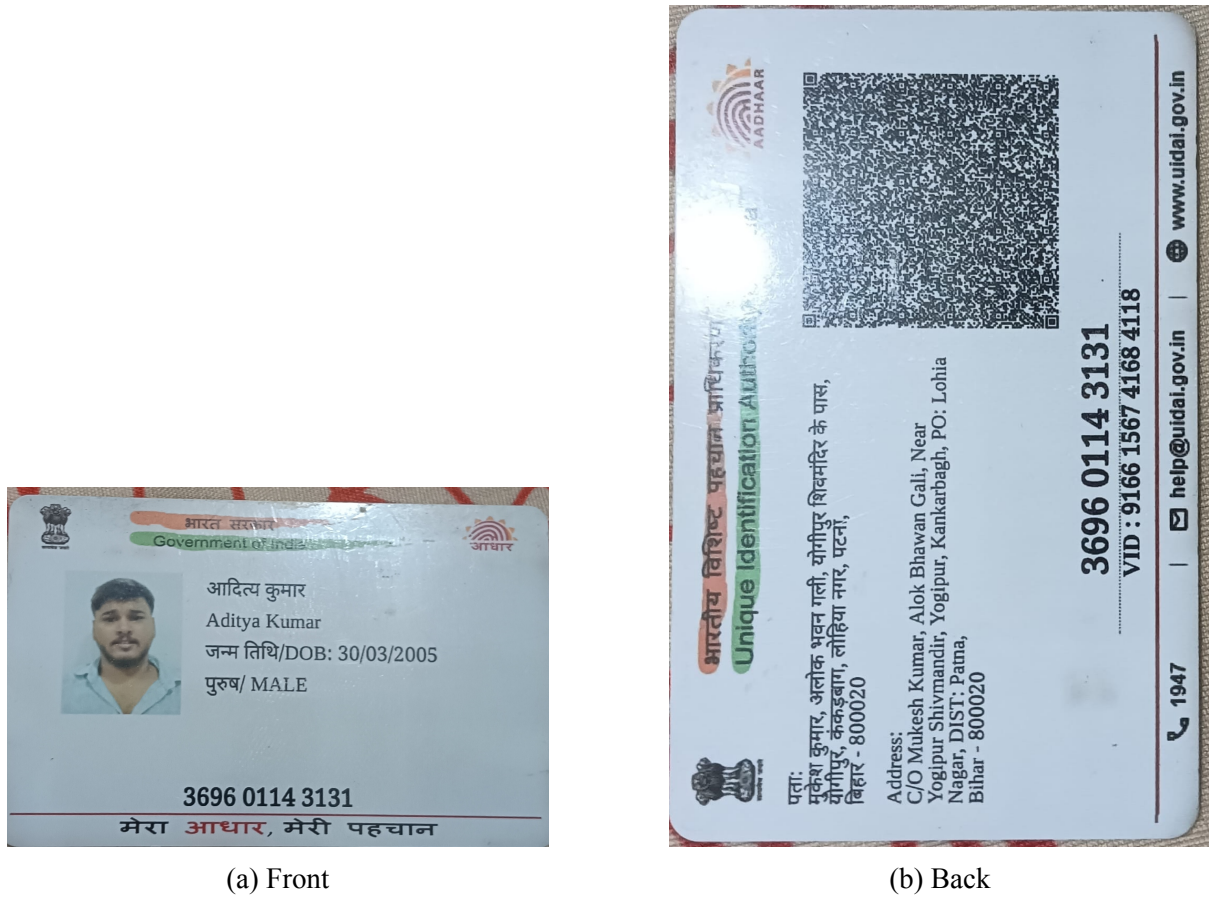


Figure 5: Real camera-captured Aadhaar images used for OCR evaluation.

Real-world images contain perspective distortion, blur, uneven illumination, reflections and cluttered backgrounds, representing practical deployment conditions.

## 2 Document Structure

Table 1: Primary Aadhaar Identifiers

| Identifier               | Description                     |
|--------------------------|---------------------------------|
| Aadhaar Number           | Unique 12-digit identifier      |
| VID                      | 16-digit Virtual ID             |
| Name                     | Cardholder name                 |
| Date of Birth            | Date of birth                   |
| Gender                   | Male/Female/Other               |
| Photograph               | Face image                      |
| Address                  | Residential address             |
| QR Code                  | Encoded demographic information |
| Enrollment Number        | Enrollment reference            |
| UIDAI Logo               | Official branding               |
| Government of India Logo | Authenticity mark               |

### 3 Identifiers and OCR Keywords

Frequently occurring keywords used for document classification and OCR field localization are:

- Government of India
- भारत सरकार
- UIDAI
- Aadhaar
- आधार
- Unique Identification Authority of India
- भारतीय विशिष्ट पहचान प्राधिकरण
- Aadhaar Number
- Your Aadhaar Number
- VID
- Virtual ID
- DOB
- Date of Birth
- Male
- Female
- Address
- Enrollment Number
- QR Code
- Mera Aadhaar Meri Pehchaan

### 4 OCR Challenges

Major OCR challenges include:

- Multiple document layouts
- Bilingual text (Hindi and English)
- Small font sizes
- QR code localization
- Camera perspective distortion
- Motion blur
- Shadows and reflections
- Uneven illumination
- Background clutter
- Compression artifacts
- Printed versus digital documents
- Privacy-preserving masked Aadhaar numbers



Table 2: Relative OCR Difficulty

| Layout                    | Difficulty |
|---------------------------|------------|
| Official e-Aadhaar        | Low        |
| PVC Aadhaar               | Medium     |
| Printed/Laminated Aadhaar | Medium     |
| Masked Aadhaar            | Medium     |
| Camera-Captured Aadhaar   | High       |